

Exploring Jupiter:

Few more secrets revealed

The largest planet in our Solar system has been in the spotlight for quite a while now. Here are some of the latest discoveries about Jupiter

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IT JUST GOT AN OFFICIAL DATE

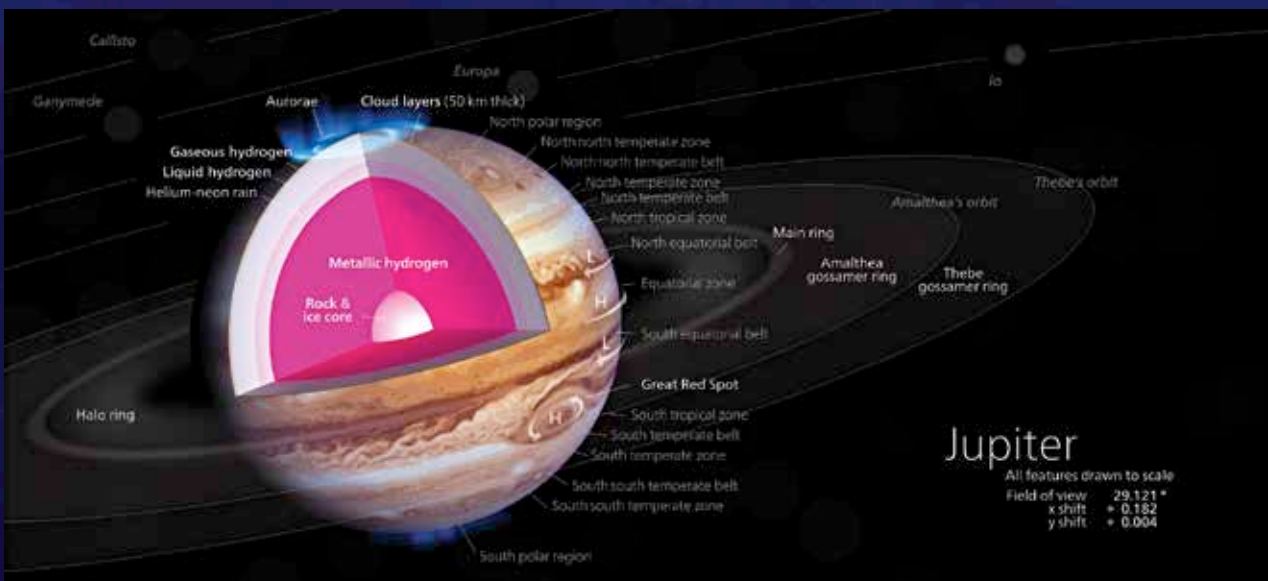
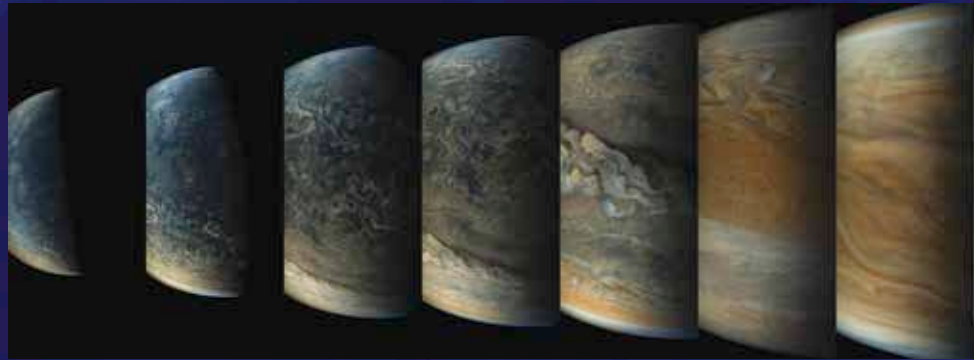
No not that kind of date. Scientists finally have figured out how old Jupiter really is – and it's pretty darn old. In no big surprise to anybody, Jupiter is also the oldest planet in our solar system. Thomas Kruijer, who headed the teams over at University of Muenster in Germany and Lawrence Livermore National Laboratory in California, published their results in the Proceedings of the National Academy of Sciences. Jupiter's solid core formed just after the formation of the Sun's nebula 4.6 billion years ago, according to the study. The gravi-

tational pull exerted by the Sun after its formation from a giant cloud of gas and dust, ended up making a 'disc' of sorts, and the leftover materials ended up making the planets. Jupiter was already 20 times as huge as Earth, just after one million years of formation (Just for comparison, 318 Earths can comfortably fit in Jupiter right now). But its growth rate slowed down and it didn't end up reaching 50 Earth masses for 2-3 million years later. The fact that it also grew so large to have a humongous gravitational field also probably accounts for the lack of planets between Mars and Jupiter. The scientists managed to figure out the exact age of the gas giant by comparing pieces of meteorites that fell to Earth

from Jupiter. These meteorites, which are made of iron also have impurities of molybdenum and tungsten. These impurities are from two distinct reservoirs that were quite a distance off and did not merge until 3-4 million years after the Solar System formation. By dating the number of isotopes of these impurities that are present in the meteorite shards, the scientists are able to find how old exactly the gas giant is. Learn more at <http://digit.in/OldJupitr>.

SENDING STUFF TO JUPITER!

After the success of our manned mission to Moon and our planned missions to Mars humankind's sights are next set on Jupiter. There have been a number



Is metallic hydrogen just hydrogen that listens to the Big Four?